

STATISTICS

Supplementary Document: Paired Populations

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Paired Populations

Statistics



Up to now, we have investigated the difference between means of two populations that are **independent**. When the populations are paired (repeated) measurements, the populations are dependent.

For example:

- The weight of people before and after a diet program
- The level of a blood parameter before and after taking a new drug
- The prices of some products sold in two different shopping centers

In the case of dependent populations such as the above examples, we should use a different technique to estimate the difference between population means.

Paired Populations

Statistics



For paired populations:

Let n paired members be chosen from two dependent populations with random variables X_1, X_2 and means μ_1, μ_2 . Then,

$$D_i = X_{1i} - X_{2i}, \quad i = 1, 2, 3, \dots, N$$

is a new random variable that represents of difference between paired members . Thus, the population mean will be

$$\mu_D = \frac{\sum_{i=1}^N D_i}{N}$$

and the sample mean and sample standard deviation will be

$$\bar{D} = \frac{\sum_{i=1}^n D_i}{n} \quad S_D = \sqrt{\frac{\sum_{i=1}^n (D_i - \bar{D})^2}{n - 1}}$$

Paired Populations

Statistics



For normally distributed populations:

$$t_{n-1} = \frac{\bar{D} - \mu_D}{S_D / \sqrt{n}}$$

The confidence interval for μ_D is: $\left[\bar{D} - t_{n-1, \alpha/2} \frac{S_D}{\sqrt{n}}, \bar{D} + t_{n-1, \alpha/2} \frac{S_D}{\sqrt{n}} \right]$

For any populations ($n > 30$)

$$z = \frac{\bar{D} - \mu_D}{S_D / \sqrt{n}}$$

The confidence interval for μ_D is: $\left[\bar{D} - z_{\alpha/2} \frac{S_D}{\sqrt{n}}, \bar{D} + z_{\alpha/2} \frac{S_D}{\sqrt{n}} \right]$

Example

Statistics



Example: To test the effect of an online tutorial on the students, 10 students are randomly chosen and their grades are measured before and after the online tutorial.

Students	After the online tutorial	Before the online tutorial
Student_1	62	61
Student_2	71	76
Student_3	66	70
Student_4	87	91
Student_5	69	46
Student_6	77	80
Student_7	45	37
Student_8	61	73
Student_9	89	90
Student_10	65	70

Find the 95% confidence interval for the difference between the mean of grades before and after the online tutorial.

Example

Statistics



If populations were independent, ...

$$n_1 = n_2 = 10$$

$$\bar{X}_2 = \frac{\sum_{i=1}^n X_{2i}}{n_2} = 69.4$$

$$\bar{X}_1 = \frac{\sum_{i=1}^n X_{1i}}{n_1} = 69.2$$

$$S_1^2 = \frac{\sum_{i=1}^n (X_{1i} - \bar{X}_1)^2}{n_1 - 1} \cong 167.29$$

$$S_2^2 = \frac{\sum_{i=1}^n (X_{2i} - \bar{X}_2)^2}{n_2 - 1} \cong 303.16$$

for unequal variances,...

$$v = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2 - 1}} \cong 16.61 \cong 17$$

$$t_v = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$t_{\frac{0.05}{2}, 17} = 2.11$$

$$\left[(\bar{X}_1 - \bar{X}_2) - t_{\frac{\alpha}{2}, v} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}, (\bar{X}_1 - \bar{X}_2) + t_{\frac{\alpha}{2}, v} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} \right] = [-14.67, 14.27]$$

INCORRECT SOLUTION !,
SINCE
POPULATIONS ARE DEPENDENT

Example

Statistics

Students	X_{1i}	X_{2i}	$D_i = X_{1i} - X_{2i}$
Student_1	62	61	1
Student_2	71	76	-5
Student_3	66	70	-4
Student_4	87	91	-4
Student_5	69	46	23
Student_6	77	80	-3
Student_7	45	37	8
Student_8	61	73	-12
Student_9	89	90	-1
Student_10	65	70	-5

$$\bar{D} = \frac{\sum_{i=1}^n D_i}{n} = -0.2$$

$$S_D = \sqrt{\frac{\sum_{i=1}^n (D_i - \bar{D})^2}{n-1}} \cong 9.60$$

$$t_{n-1} = \frac{\bar{D} - \mu_D}{S_D / \sqrt{n}} \quad t_{n-1, 0.05/2} = 2.26$$

$$\left[\bar{D} - t_{n-1, \alpha/2} \frac{S_D}{\sqrt{n}}, \bar{D} + t_{n-1, \alpha/2} \frac{S_D}{\sqrt{n}} \right]$$

$$= [-7.09, 6.67]$$

CORRECT SOLUTION